PRACTICAL:1

Aim :

Theory :

Code :

```
% Experiment 1
%%  Matrix Manipulation
A = [1, 2, 3; 4, 5, 6; 7, 8, 9];
% Accessing elements
element = A(2, 3);  % Gets element at row 2, column 3 (value: 6)
% Accessing rows and columns
row2 = A(2, :);     % Gets second row: [4, 5, 6]
col3 = A(:, 3);     % Gets third column: [3; 6; 9]
% Submatrices
subA = A(1:2, 2:3); % Gets 2×2 submatrix: [2, 3; 5, 6]
% Append
B = [A; [10, 11, 12]];  % Vertical
C = [A, [13; 14; 15]];  % Horizontal
%% Matrix Functions
A = [1, 2; 3, 4];
det_A = det(A);           % Determinant
inv_A = inv(A);           % Inverse
[V, D] = eig(A);          % Eigenvalues and eigenvectors
[U, S, V] = svd(A);       % Singular value decomposition
rank_A = rank(A);         % Rank
trace_A = trace(A);       % Trace (sum of diagonal elements)
t = A'; % Transpose
%% Creating Matrices
% Direct assignment
A = [1, 2, 3; 4, 5, 6; 7, 8, 9];
% Using functions
B = zeros(3);      % 3×3 matrix of zeros
C = ones(2, 4);    % 2×4 matrix of ones
D = eye(3);        % 3×3 identity matrix
E = rand(2, 3);    % 2×3 matrix of random values (0 to 1)
F = randn(3, 2);   % 3×2 matrix of normally distributed random values
%% Colon operator for creating vectors
x = 1:5;        % Creates [1, 2, 3, 4, 5]
y = 1:0.5:3;    % Creates [1, 1.5, 2, 2.5, 3]
```

Output:

Name	Value	Size	Class
A	[1,2,3;4,5,6;7,8,9]	3×3	double
B	[0,0,0;0,0,0;0,0,0]	3×3	double
C	[1,1,1,1;1,1,1,1]	2×4	double
col3	[3;6;9]	3×1	double
D	[1,0,0;0,1,0;0,0,1]	3×3	double
det_A	-2	1×1	double
E	[0.8147,0.1270,0.6324;0.9058,0.9134,0.0975]	2×3	double
element	6	1×1	double
F	[-0.4336,2.7694;0.3426,-1.3499;3.5784,3.0349]	3×2	double
inv_A	[-2,1;1.5000,-0.5000]	2×2	double
rank_A	2	1×1	double
row2	[4,5,6]	1×3	double
S	[5.4650,0;0,0.3660]	2×2	double
subA	[2,3;5,6]	2×2	double
t	[1,3;2,4]	2×2	double
trace_A	5	1×1	double
U	[-0.4046,-0.9145;-0.9145,0.4046]	2×2	double
V	[-0.5760,0.8174;-0.8174,-0.5760]	2×2	double
x	[1,2,3,4,5]	1×5	double
y	[1,1.5000,2,2.5000,3]	1×5	double

Conclusion: